

CHE 565 -- Homework 3

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Problem 1

Setup

Function :

$$f(x_1, x_2) = 0.5x_1^2 + 1.0x_1x_2 - x_1 + 1.0x_2^2 - x_2$$

At $x_0 = [1, 1]$, $f(x_0) = 0.5$

Gradient of f :

$$\begin{bmatrix} 1.0x_1 + 1.0x_2 - 1 \\ 1.0x_1 + 2.0x_2 - 1 \end{bmatrix}$$

At $x_0 = [1, 1]$, $Df(x_0) = \begin{bmatrix} 1. \\ 2. \end{bmatrix}$

Function g :

$$\begin{bmatrix} g_1(\alpha) \\ g_2(\alpha) \end{bmatrix} = \begin{bmatrix} -\alpha(1.0x_1 + 1.0x_2 - 1) + x_1 \\ -\alpha(1.0x_1 + 2.0x_2 - 1) + x_2 \end{bmatrix}$$